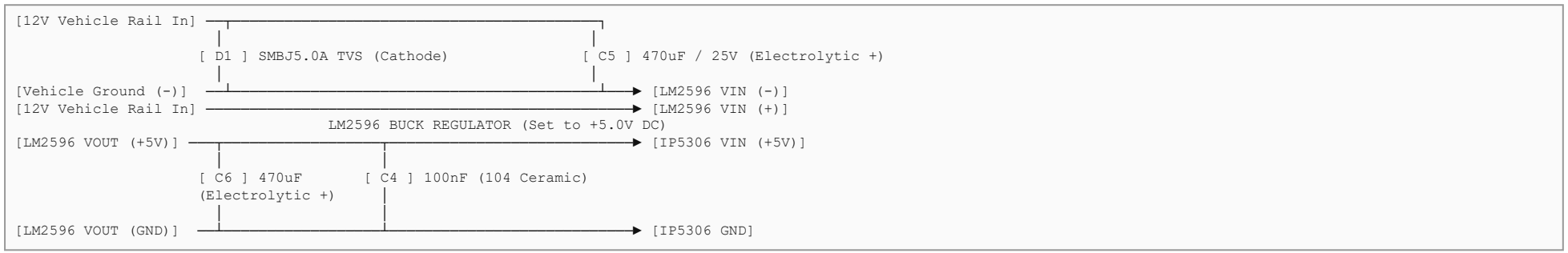
HARDWARE ARCHITECTURE & SUBSYSTEM SPECIFICATION (SHEET 1)

System Objective: Ultra-low-cost, battery-buffered automotive vision streaming prototype using Luckfox Pico Mini B (RV1103 SoC, 64MB DDR2, 128MB SPI NAND) and SmartSens SC3336 3MP sensor.

1. FUNCTIONAL SUBSYSTEM OVERVIEW

SUBSYSTEM	COMPONENT SELECTION	ENGINEERING SPECIFICATION
Vision Processor	Luckfox Pico Mini B (RV1103 SoC, 64MB DDR2, 128MB SPI NAND)	ARM Cortex-A7 @ 1.2GHz + RISC-V co-processor, 64MB DDR2, built-in ISP 3.2, HW H.264/H.265 VPU.
Optical Sensor	SmartSens SC3336 3MP MIPI CSI-2	2304x1296 @ 30fps, 2-lane MIPI D-PHY, 98.3 Deg wide-angle lens, high SNR low-light automotive profile.
Non-Volatile Storage	128MB SPI NAND Flash (Onboard)	Zero SD card topology. Houses U-Boot, Linux kernel, Busybox rootfs, and RKMPI streamer daemon.
Wireless Network	Realtek RTL8188EUS USB Dongle	802.11b/g/n 2.4GHz SoftAP mode, 150Mbps PHY rate, native in-kernel '8188eu' driver.
Power Management	Injoinic IP5306 Synchronous PMIC	2.1A CC/CV LiPo charger, 5.0V @ 2.4A boost output, autonomous load-sharing, over/under voltage protection.
Input Pre-Regulator	LM2596 DC-DC Step-Down Buck	Steps down vehicle electrical rail (9V-16V nominal, transients up to 40V) to regulated 5.0V DC.

2. POWER CONDITIONING & PROTECTION TOPOLOGY



ORGANIZATION / PROJECT DASHCAM PROTOTYPE PROJECT	SIZE A4	DOCUMENT NUMBER ENG-DASH-HW-001	DATE 2026-10-08	SHEET 1 OF 2	
DOCUMENT TITLE HARDWARE SPECIFICATION SHEET	REV 1.4	DESIGN ENGINEER Souvik H	ENGINEER REGISTRATION NO. IN-ENG-2022-22824		
		ENGINEERING DISCIPLINE EMBEDDED SYSTEMS & HARDWARE	STATUS RELEASED		

HARDWARE ARCHITECTURE & SUBSYSTEM SPECIFICATION (SHEET 2)

3. HARDWARE INTERFACE & SIGNAL SPECIFICATION MATRIX

INTERFACE / BUS	SOURCE -> DESTINATION	PHYSICAL STANDARD	BAUD RATE / BANDWIDTH	ENGINEERING PURPOSE
MIPI CSI-2	SC3336 -> RV1103	Differential D-PHY (2-lane)	800 Mbps / lane	Direct transport of raw 10-bit Bayer pixel stream to ISP.
SCCB Control	RV1103 <-> SC3336	I2C Protocol (0x3C)	400 kHz Fast Mode	Sensor initialization, exposure, white balance, and gain control.
SPI Flash Bus	RV1103 <-> SPI NAND	Standard Quad SPI	50 MHz to 100 MHz	Execution interface for bootloader, Linux kernel, and rootfs.
USB 2.0 Host	RV1103 <-> RTL8188EUS	High-Speed USB 2.0 PHY	480 Mbps	Network physical interface transporting H.264 RTSP video stream.
Status Indicator	RV1103 -> Status LED	GPIO Out via R4 (470 Ohm)	DC Digital	Visual heartbeat and streaming state indication.
Master Switch	Push Button -> RV1103	GPIO Active Low (R2: 10k)	DC Digital	System mode selection and safe shutdown control switch.

4. PASSIVE PROTECTION SCHEDULE

REF DES	COMPONENT VALUE	PACKAGE	FUNCTIONAL LOCATION & ENGINEERING PURPOSE
D1	SMBJ5.0A (600W)	SMB / DO-214AA	12V vehicle input line clamp; shunts inductive kickback and alternator load dump spikes.
C5, C6	470uF / 25V Radial	Aluminum Electrolytic	C5 filters vehicle input ripple; C6 buffers LM2596 output during Wi-Fi transmission current spikes.
C1-C4	100nF (104) 50V	Ceramic Radial / 0805	High-frequency decoupling placed directly at IP5306 output, SBC power port, Wi-Fi port, and buck node.
R1	100 Ohm / 0.25W	Metal Film Axial	IP5306 keep-alive dummy load: maintains 50mA static pull to prevent automatic power bank shutdown.
R2, R3	10 kOhm / 0.25W	Carbon Film Axial	Pull-up resistor for power switch and pull-down resistor for boot state latching.
R4, R5	470 Ohm, 1 kOhm	Metal Film Axial	R4 limits status LED current; R5 ties USB ID pin to ground securing default USB Host mode.

ORGANIZATION / PROJECT DASHCAM PROTOTYPE PROJECT	SIZE A4	DOCUMENT NUMBER ENG-DASH-HW-001	DATE 2026-10-08	SHEET 2 OF 2
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